STAT 6611: Theory & Application of Linear Models

1 Preliminaries: Generalized Inverses/Projection Matrices

Theorem 1.1. *For any $m \times n$ matrix A of rank $r \geq 1$, there exist nonsingular matrices P (size $m \times m$) and Q (size $n \times n$) such that*

$$PAQ = \begin{bmatrix} D_{r\times r} & 0_{r\times(n-r)} \\ 0_{(m-r)\times r} & 0_{(m-r)\times(n-r)} \end{bmatrix},$$

where $D \equiv D_{r\times r}$ is a nonsingular $r \times r$ diagonal matrix.

The $n \times m$ matrix

$$G = Q \begin{bmatrix} D^{-1} & F \\ H & B \end{bmatrix} P$$

satisfies $AGA = A$, where F, H, and B are any matrices of appropriate dimensions.

Proof. Note that

$$PAQ = \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix} \iff A = P^{-1} \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix} Q^{-1}.$$

Let

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \vdots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} = (\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_n)$$

represent an $m \times n$ matrix of rank r $(r \geq 1)$.

Take K_1 to be an $m \times r$ matrix whose columns are r linearly independent columns of A, say

$$K_1 = (\underline{a}_{s_1}, \ldots, \underline{a}_{s_r}).$$

out of n columns of A, pick r of them that are lin. independt

Every column of A must then be a linear combination of the columns of K_1.

the r columns of A given by K_1 are linear combinations of K_1, but the other n-r columns of A must be linear combinations of K_1 columns too. (if not rank(A) > r holds)

In other words, for some $t_{11}, \ldots, t_{r1}, \ldots, t_{1n}, \ldots, t_{rn}$,

each column of A (n of these)

$$\underline{a}_1 = t_{11}\underline{a}_{s_1} + \cdots + t_{r1}\underline{a}_{s_r} = (\underline{a}_{s_1}, \ldots, \underline{a}_{s_r}) \begin{pmatrix} t_{11} \\ \vdots \\ t_{r1} \end{pmatrix} = K_1 \underline{t}_1,$$

$$\vdots$$

$$\underline{a}_n = t_{1n}\underline{a}_{s_1} + \cdots + t_{rn}\underline{a}_{s_r} = (\underline{a}_{s_1}, \ldots, \underline{a}_{s_r}) \begin{pmatrix} t_{1n} \\ \vdots \\ t_{rn} \end{pmatrix} = K_1 \underline{t}_n.$$

Hence,

$$A = (\underline{a}_1, \ldots, \underline{a}_n) = (K_1\underline{t}_1, \ldots, K_1\underline{t}_n) = K_1(\underline{t}_1, \ldots, \underline{t}_n) = K_1 T_1$$

m x n; m x r; r x n

for some $r \times n$ matrix $T_1 = (\underline{t}_1, \ldots, \underline{t}_n)$.

It turns out that $\text{rank}(T_1) = r$. = rank(A) = rank(K_1)

To see this, choose r rows of A which are linearly ind. & put into a r x n matrix $\tilde{A}$

And choose corresponding rows of K_1 & put into a matrix $\tilde{K}_1$

Then $\tilde{A} = \tilde{K}_1 T_1$ (r x n = r x r · r x n)

It must be that $\tilde{K}_1$ has rank r. (otherwise, rank($\tilde{A}$) < r)

then $\text{rank}(T_1) = \text{rank}(\tilde{K}_1^{-1}\tilde{A}) = \text{rank}(\tilde{A}) = r.$

Key insight now: $A = K_1 T_1$ ($m \times n$ $=$ $m \times r$ $r \times n$) both K_1 & T_1 have $\text{rank}(A) = r$ ($r \le \min\{m, n\}$)

Next, take K to be an $m \times m$ nonsingular matrix and T to be an $n \times n$ nonsingular matrix of the forms

$$K = (K_1, K_2) \qquad T = \begin{pmatrix} T_1 \\ T_2 \end{pmatrix}$$

(K: $m \times m$; K_1: $m \times r$, K_2: $m \times (m-r)$; T: $n \times n$; T_1: $r \times n$, T_2: $(n-r) \times n$)

add $n-r$ rows in $\mathbb{R}^n$ independent of rows of T_1

add $(m-r)$ columns in $\mathbb{R}^m$ independent of each other & columns of K_1

Observe that

$$A = K_1 T_1 = (K_1, 0) \begin{pmatrix} T_1 \\ T_2 \end{pmatrix} = (K_1, K_2) \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \begin{pmatrix} T_1 \\ T_2 \end{pmatrix} = K \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} T,$$

(read this way; $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$: $m \times n$; def; $\begin{pmatrix} T_1 \\ T_2 \end{pmatrix}$: $m \times n$)

where I_r is the $r \times r$ identity matrix.

Hence,

$$K^{-1} A T^{-1} = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix},$$

(P, A, Q, D)

and $P = K^{-1}, Q = T^{-1}$, and $D = I_r$ satisfy the requirements.

Finally, given the existence of P, Q, and D, we have

(A, G, A)

$$\begin{aligned} AGA &= P^{-1} \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix} Q^{-1} Q \begin{bmatrix} D^{-1} & F \\ H & B \end{bmatrix} P P^{-1} \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix} Q^{-1} \\ &= P^{-1} \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} D^{-1} & F \\ H & B \end{bmatrix} \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix} Q^{-1} \\ &= P^{-1} \begin{bmatrix} I_r & DF \\ 0 & 0 \end{bmatrix} \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix} Q^{-1} = P^{-1} \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix} Q^{-1} = A. \end{aligned}$$

□

Definition. A generalized inverse of a matrix A is any matrix G that satisfies $AGA = A$.

Notation: For any matrix A, we use A^- to denote an arbitrary generalized inverse of A. G is a generalized inverse of A (i.e. AGA=A) we write $G=A^-$

Theorem 1.1 indicates that:

- any matrix A has a generalized inverse

- if $A = 0$ is the $m \times n$ zero matrix, then any $n \times m$ matrix G is a generalized inverse of A. any matrix G (nxm)

$AGA = 0G0 = 0 = A$

- A has infinitely many generalized inverses unless A is nonsingular

can pick H, F, B as arbitrary

- if A is nonsingular, it has a unique generalized inverse, which is A^{-1}.

i.e. A must be square $m=n=r=\text{rank}(A)$

$PAQ = D_{rxr}$ $A^- = A^{-1} = QD^{-1}_{rxr}P$

- Also, Theorem 1.1 suggests an algorithm for computing a generalized inverse.

- Note in Theorem 1.1 that values of P, Q, D are not unique.

$PAQ = \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix}$ see previous example